

Assignment №1- color image and histograms

Evaluation:

Environment set up	10%
Images saved in the correct format	15%
Upscaling/downscaling	10%
Drawing shapes	10%
Video annotation	10%
Histogram equalization	15%
Report	30%

THIS ASSIGNMENT WILL BE 10% OF FINAL GRADE

Setting up environment

Find instructions in Moodle for setting up the environment.

To check that everything is correct:

- create a new script
- import OpenCV, NumPy, TensorFlow and matplotlib libraries
- show their versions

```
import pandas as pd
print(pd.__version__)
```

Preparing images for this course (optional)

You can obtain images by photographing the necessary objects with your smartphone (best practice because you'll be the author of these images). If you have problems taking pictures, you can source them from free image stocks, but be careful about licensing issues. If attribution is requested, add a link to the author even if you show these images in class. Please avoid using images from Google searches or random websites; they may be someone's property. List of necessary images:

- Your selfie
- Selfie with friends and/or family
- Picture of animal (your pet, street cat, any animal from the zoo, etc.)
- Words on the paper (write several by hand and take a picture)
- Cars on the street

It is important! Store another copy of your images somewhere away from the project folder. It is best practice to have copies both in the cloud and on your local drive.

Opening images in Python, formatting them, saving formatted images from python

Create a new Python script, do all necessary imports (refer to our tutorials). It is Ok to use `image_common_utils` module if you want.

Download your images in Python and display them with matplotlib and opencv.

Choose one image and change its color scheme.

Save it. Repeat with different color schemes, and file formats. Mention in the file name what color scheme was used.

	RGB	CMYK	HEX	Grayscale	HSL
.tiff					
.jpeg					
.png					
.gif					

*Be aware not every file format supports all color schemes.

Upscaling and downscaling your images

- Try to upscale and downscale one of your images.
- Save results in separate files and compare their sizes.
- How the size effect the image quality?

Drawing shapes

Write a script that displays the image and allows the user to click anywhere on the image. Add functionality to allow the user to click twice to define two corners of a rectangle.

Draw the rectangle on the image based on the selected points.

Don't forget to add keys to close the window (usually ESC or q).

Video Annotation

Write a script to annotate each frame of the video horse.mp4 with its average of green band at the bottom-right corner. `#green_channel = frame[:, :, 1]` # Extract the green channel

Save the annotated video as horse_annotated.mp4.

Histogram equalization

- Present the Cumulative histogram vs. histogram of your image for each of the three bands.
- Apply thresholding to your image in dark or bright areas and present the results.
- Select transformation on the image, for example, Log transform.
- Apply **Contrast Limited Adaptive Histogram Equalization (CLAHE) or other Histogram Equalization approach** to equalize images and present the results.
- Contrast enhancement by Histogram equalization

NB! Whenever you want to apply any transformation on multiband image you need to split RGB to channels and treat each channel independently. After the transformations are applied you need to stack the channels back to RGB.

Important: Transformations performed on a grayscale image wouldn't count for this task. You need to apply transformations on color image.

What to submit?

PDF with:

1. List of library versions.
2. Comparison of the sizes and quality of different image files.
3. Short report (not more than one-page A4, 11pt) on the advantages and disadvantages of different formats and color schemes.
4. Table with color scheme/file format with short comment in each cell. Screenshot of folder with files in different image and color formats.
5. Screenshot of folder with upscaled/downscaled images (with sizes of files) and short report about quality changes.
6. Picture with your shapes drawn
7. Screenshots of 3 different frames from your video
8. Histograms for each band and cumulative histogram before transformation
9. Image before and after transformation + histograms after transformation

Your code

Please write detailed comments to your code so I can understand it.

IMPORTANT!!!

I am not a Hebrew speaker, please submit your assignments in English

1. Add all images, tables and text to a **single** .docx or .pdf (please keep reasonable resolution of images)
2. I expect to receive only **two files** from each of you: **report** (.pdf or .doc or .docx) and **code** (.py or .jpynb)
3. Add your name to the name of the report and code

Example of naming:

laroslav_Grozlov_report_hw1.pdf

laroslav_Grozlov_code_hw1.py

Name_Surname_code_hw1.pdf

DON'T SEND YOUR IMAGES AND VIDEOS!

Moodle have limit of 50mb for uploads so to avoid it I prefer to receive your assignments on **Google Drive.**

Link for submission:

<https://forms.gle/MTatmEU7LKMPYsHZ8>